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A Synthesis of Australian Cenozoic Palaeomagnetic Results

M. W. McElhinny, B. J. J. Embleton and P. Wellman*

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Summary

Large numbers of K–Ar determinations have now been made on the Cenozoic volcanic rocks of eastern Australia. These age determinations make it possible to synthesize the palaeomagnetic results from 357 lava flows studied to date by dividing them into six age groups ranging from 60 My to Recent. We use the criterion that palaeomagnetic poles with VGP co-latitudes $> 40°$ are intermediate between normal and reversed, as has been demonstrated in Iceland. Overall 11–15 per cent of flows are classified as intermediate in this way, whereas only 3 per cent of flows with ages < 5 My and 20–30 per cent in the time range 35–25 My are intermediate. The observations from the younger group are consistent with that to be expected from the known polarity time-scale for the last 5 My. Those from the older group suggest the frequency of reversals was much greater than had previously been supposed during the mid-Tertiary. The apparent polar wander path derived from these results shows a distinct zig-zag with a westerly excursion around 30 My and a knee around 18 My.

1. Introduction

Throughout the Cenozoic widespread vulcanism took place along the entire length of eastern Australia. Early palaeomagnetic investigations of some of these volcanics (Irving & Green 1958; Mumme 1963; Irving Stott & Ward 1961) were hampered by the lack of precise age control, which was derived largely from general geomorphological considerations or limited fossil evidence. Wellman, McElhinny & McDougall (1969) combined the palaeomagnetic investigations with K–Ar dating, and this enabled some detail to be made of the Cenozoic apparent polar wander path for Australia.

A very thorough study has now been made of the whole pattern of Cenozoic vulcanism in eastern Australia using K–Ar dating (Wellman & McDougall 1973). These K–Ar determinations show that much of the early palaeomagnetic work (Irving *et al.* 1961; Mumme 1963) was carried out on volcanics whose ages ranged from Palaeocene (60 My) to Early Miocene (17 My). Consequently it has been very difficult to compare such data with the more recent results, each of which refer to a much more restricted time range.

Of all the volcanics in eastern Australia and New Guinea that have now been studied for palaeomagnetic purposes using standard cleaning techniques, 357 lava flows show stable directions of magnetization. Some small studies not previously

* Now at Bureau of Mineral Resources, Geology and Geophysics, Canberra, A.C.T.

reported in the literature are presented below. All the flow directions of magnetization have then been regrouped according to age into six groups to provide a complete synthesis of the results. The results of Green & Irving (1958) from four sites in Tasmania are still of unknown age and are not discussed further.

2. Results of some small studies

Fig. 1 shows the various volcanic provinces in eastern Australia that have been studied for palaeomagnetic purposes. Detailed studies involving standard cleaning techniques have already been reported from Nandewar, Liverpool and Barrington Volcanoes (Wellman *et al.* 1969), from Tweed Volcano and the Main Range Province (Robertson 1966; Wellman 1973) and from the Older (Mumme 1963) and Newer (Aziz-ur-Rahman 1971) Volcanics of Victoria. Facer (1968) reports results from two

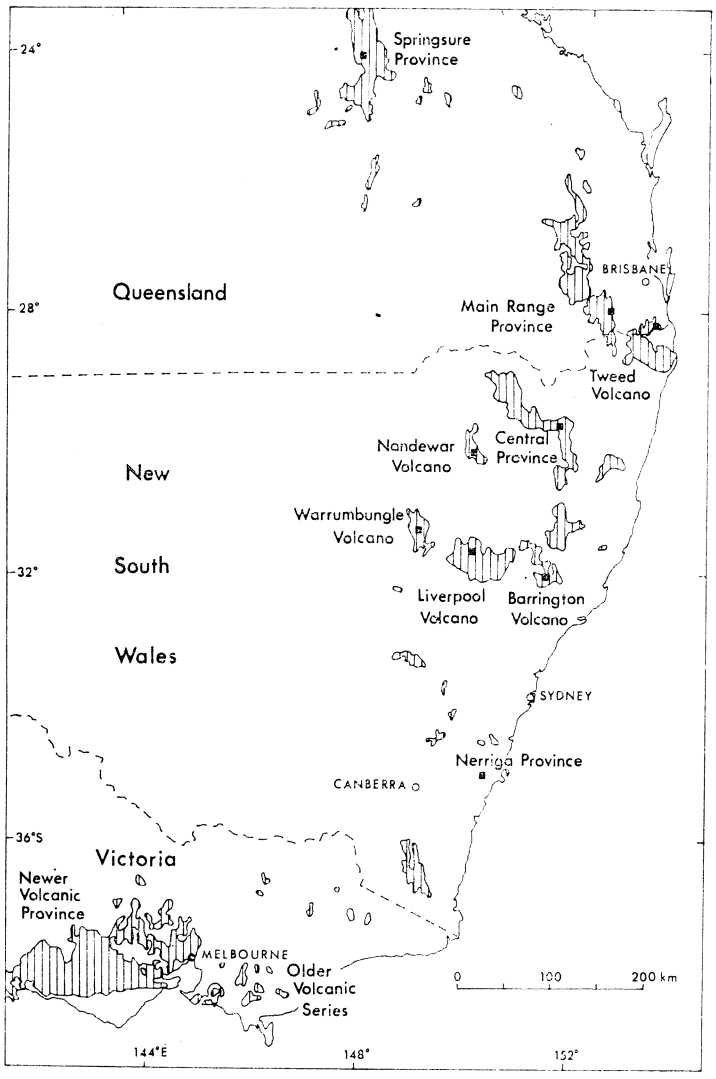


FIG. 1. Distribution of Cenozoic volcanics in eastern Australia. The areas sampled for palaeomagnetism are named.

flows from Mt Canoblas near Orange, N.S.W., whose age is now known to be about 11 My. The directions were scattered and differed markedly from the axial dipole field direction. Results from some of the other volcanic provinces not previously reported are described briefly below.

(a) *Central Province*

In the eastern part of the Central Province in northern N.S.W. a section of 13 flows was sampled from the Gwydir Highway on the western slopes of Waterloo Range, 12 km west of Glen Innes. After magnetic cleaning 31 samples collected from these flows give the results listed in Table 1. Four polarity changes are observed in

Table 1

Flow-mean palaeomagnetic data obtained after magnetic cleaning. The flows from each region are entered in strict stratigraphic order, youngest at the head of the table.

(a) Central Province (Waterloo Range); 151·5° E, 30° S. 154 m of the succession are represented.

Flow No.	*N*	*D*°	*I*°	Long.	Lat.	Polarity
13	2	191	−48	343° E	30° S	*I*
12	3	180	−59	332 E	20 S	*I*
11	3	013	−66	127 E	69 S	*N*
10	3	012	−79	144 E	51 S	*N*
9	2	105	+59	209 E	29 S	*I*
8	2	157	−70	318 E	04 S	*I*
7	2	069	−55	085 E	35 S	*N*
6	1	049	−61	092 E	49 S	*N*
5	2	128	+62	209 E	47 S	*I*
4	3	220	+29	049 E	51 S	*R*
3	3	002	−79	150 E	52 S	*N*
2	2	325	−60	210 E	60 S	*I*
1	3	309	−55	221 E	47 S	*I*

(b) Nerriga Province—a summary of the data is provided in the text, see Section 2(b).

(c) Warrumbungle Volcano; 149·1° E, 31·3° S.

Flow No.	*N*	*D*°	*I*°	Long.	Lat.	Polarity
Parramatta Mountain (123 m of the succession are represented)						
12	2	192	+35	014° E	74° S	*R*
11	4	222	+33	050 E	50 S	*R*
10	3	172	+30	302 E	73 S	*R*
9	1	023	−05	010 E	54 S	*N*
8	3	313	−45	233 E	49 S	*I*
7	2	176	−75	147 E	02 S	*I*
6	3	343	−51	233 E	76 S	*N*
Exmouth Mountain (160 m of the succession are represented)						
5	3	211	+37	046 E	60 S	*R*
4	3	181	+36	325 E	79 S	*R*
3	3	151	−07	285 E	46 S	*I*
2	3	160	+07	291 E	56 S	*I*
1	3	349	−65	173 E	72 S	*N*

(d) Springsure Province (Mount Boorambool); 148° E, 24° S. 261 m of the succession are represented.

Flow No.	*N*	*D*°	*I*°	Long.	Lat.	Polarity
7	1	055	−53	083° E	41° S	*N*
6	1	114	−53	278 E	05 S	*I*
5	2	114	−64	109 E	02 S	*I*
4	2	045	−61	095 E	48 S	*N*
3	2	337	−32	251 E	67 S	*I*
2	1	194	+43	064 E	77 S	*R*
1	2	306	+64	293 E	05 S	*I*

the sequence and directions from three of the flows which depart widely from the others are regarded as transitional. Three K–Ar whole rock determinations from the base and top of the lava sequence in the surrounding area give ages in the range 34–33 My, so the lavas are regarded as Early Oligocene in age.

(b) *Nerriga Province*

Small areas of basaltic flows occur in south-eastern N.S.W. near the settlement of Nerriga (site co-ordinates 150·1°E, 35·1°S). The thickest section in the valley of the Endrick River is about 120 m thick. Two flows from near the top of this sequence gave whole rock K–Ar ages that agree within experimental error at 45·6±1·5 My; the flows are of Middle Eocene age. Twenty flows were sampled in this section and NRM directions, which remained essentially unchanged up to 450 Oe, were all normally magnetized with steep dips. On thermal demagnetization in 200–325°C the directions of test samples from 12 flows formed a very tight group. The lack of reversals and the tight grouping suggests the flows were extruded over a very short period of time. The mean direction $D = 077°$, $I = -75°$ ($R = 11{\cdot}948$, $k = 211$, $\alpha_{95} = 3°$) of the 12 thermally cleaned flows is regarded as representing only a single point in time. The corresponding pole position lies at 115°E, 36°S.

(c) *Warrumbungle Volcano*

This volcano lies on the western margins of the Eastern Highlands. Two sections of lava flows were investigated. In the western central part at Mount Exmouth (Mount Wambelong) there is a 700-m section of about eight flows. Whilst flows in this section have not been dated, similar flows 10 km to the east at Woorut Mountain gave whole rock K–Ar ages in good agreement at 15·6±0·4 My, and a similar flow 10 km to the west gave an age of 14·5±0·5 My. On the southern margin of the volcanic cone, 18 km to the south, a 130-m section of eight flows is exposed on the slopes of Parramatta Mountain. Two flows near the top of the section gave whole rock K–Ar ages agreeing within experimental error at 14·4±0.4 My.

Fifteen flows were sampled from these two sections, three of which gave inconsistent within flow sample directions after magnetic cleaning. Results from the remaining 12 flows are listed in Table 1. In both sections the base consists of a normally magnetized flow followed by two flows that could be regarded as being part of a transition zone to the overlying reversely magnetized flows.

(d) *Springsure Province*

In central eastern Queensland is a thick sequence of mainly basaltic lavas forming the Springsure Province. The basal flows of olivine basalt with a total thickness of 70 m are overlain by a 250-m section of basalts and trachytes, intruded by trachyte and rhyolite. K–Ar ages reported by Webb & McDougall (1967) show the lower flows are about 33 My old, whilst the upper flows and intrusions are 27·0±0·5 My old.

Samples were collected from seven flows in the 275-m section between the town of Springsure and Mount Boorambool. After magnetic cleaning the flow mean directions are those listed in Table 1. The section is interpreted as consisting of reverse and transitional flows near the base and normal and transitional flows near the top.

3. Cenozoic synthesis

In Table 2 are summarized the palaeomagnetic studies of Cenozoic volcanics made in eastern Australia and New Guinea together with the range of K–Ar ages associated with them. The 357 flows giving stable directions of magnetization have been regrouped into time ranges as in Table 3 from the known or inferred ages from K–Ar

Table 2

Summary of palaeomagnetic studies of Cenozoic volcanics in eastern Australia and New Guinea. N *or* R *means VGPs with co-latitudes* $\leqslant 40°$, I *means those* $> 40°$.

Volcano or Province	Reference	K–Ar ages	Flows (*N* or *R*)	Flows *I*
Older Volcanics, Victoria	Mumme (1963)	60–25	13	7
Barrington Volcano	Wellman *et al.* (1969)	52	33	2
Tertiary basalts	Irving *et al.* (1961)	50–17	6	2
Nerriga Province	This paper	46	12	—
Liverpool Volcano	Wellman *et al.* (1969)	34	38	10
Central Province	This paper	34–33	6	7
Springsure Province	This paper	33–27	3	4
Tweed Volcano	Wellman (1973)	22–20	42	—
Main Range Province	Wellman (1973)	22–20	19	3
Queensland	Robertson (1966)	24–20	14	3
Nandewar Volcano	Wellman *et al.* (1969)	18–17	41	8
Warrumbungle Volcano	This paper	16–14	8	4
Mt Canoblas	Facer (1968)	11	—	2
Newer Volcanics, Victoria	Aziz-ur-Rahman (1971)	5–0	47	1
New Guinea	Manwaring (1971)	<0·1	21	1
	Total flows	357	303	54

Table 3

Australian Cenozoic results. (T = *Total number of flows;* N = *no. used in calculating mean;* I = *no. of VGPs lying* $> 40°$ *from mean;* I′ = *no. of VGPs lying* $> 50°$ *from mean;* R = *resultant of* N *unit vectors;* K = *precision parameter;* A_{95} = *radius of circle of* 95 *per cent confidence*).

Time range (My)	T	N	I	I'	R	K	A_{95}	Lat. °S	Long. °E
60–40	55	46	9	5	43·3838	17·2	5·2	68·5	130·9
35–25	74	52	22	15	49·7346	22·5	4·3	68·9	92·4
25–20	81	75	6	6	70·6622	17·1	4·1	77·4	110·9
20–15	64	51	13	11	48·8701	23·5	4·2	79·4	66·0
5–0	47	46	1	1	—	—	1·9	86·6	86·3
<0·1	22	22	1	1	—	—	—	86	150

Sources:

60–40 My includes the Barrington Volcano, Nerriga Province, sites 1–5 of Irving *et al.* (1961) and sites 8–21 of Mumme (1963).

35–25 My includes the Liverpool Volcano, Central Province, Springsure Province and sites 1–7 of Mumme (1963).

25–20 My includes the Tweed Volcano, Main Range Province and sites A–0, N–S of Robertson (1966).

20–15 My includes Nandewar Volcano, Warrumbungle Volcano and sites 6–8 of Irving *et al.* (1961).

5–0 My is the Newer Volcanics of Victoria (Aziz-ur-Rahman, 1971).

<0·1 My is the New Guinea basalts (Manwaring 1971).

dating. We have left unaltered the separate studies of Aziz-ur-Rahman (1971) on the Newer Volcanics of Victoria and of Manwaring (1971) on Recent volcanics in New Guinea. The remaining flows ranging in age from 15 to 60 My have been arranged into four convenient groups as in Table 3.

(a) *Geomagnetic variations*

A significant feature of all except the two studies of very young volcanics is the large number of flows having widely scattered directions of magnetization away from the main group of directions. Some of these can be identified as belonging to transitional directions observed during a polarity transition (Wellman *et al.* 1969), although in some cases sudden departures occur even within part of a sequence that exhibits no polarity change. These correspond to the systematic deviations of Lawley (1970) and together with the transitional flows have been generally referred to as anomalous by Dagley *et al.* (1967). Wilson *et al.* (1972a) have made a detailed analysis of intensities of magnetization observed in about 1400 Icelandic lava flows. They demonstrate that there is a sharp decrease in intensity for flows whose VGP co-latitudes are greater than 40°. This phenomenon therefore provides a natural definition of normal and reversed polarity and of flows that can be classified as intermediate. In analysing the data for the four time ranges between 15 and 60 My, we have used this definition as a means of eliminating intermediate directions from the analysis.

Pole positions were calculated for all lava flows irrespective of whether the author regarded these as anomalous or transitional. After rejecting the grossly deviating (say > 60°) pole positions from each group, a mean was calculated and all flows with VGPs lying greater than 40° from this mean pole position were rejected. A new mean was then calculated and the process repeated until ultimately the population of poles for each group lay within 40° of their mean. All poles rejected were then regarded as intermediate or transitional. The distribution of the flows according to this classification is given in Table 2 for each individual study and in Table 3 for the overall groups.

One problem that arises is that the mean pole position from which the 40° limit is drawn is only known to an accuracy of 4–5°. Consequently there is a possible tendency to overestimate the number of intermediate VGPs. A conservative estimate may be obtained by regarding only those VGPs lying greater than 50° from the mean as being intermediate. These numbers are also given in Table 3. Of the 357 flows between 41 and 54 (minimum and maximum estimates) have intermediate VGPs or 11–15 per cent. This may be compared with 72 out of about 900 flows (8 per cent) observed by Dagley *et al.* (1967) and 18 per cent of flows reported by Wilson *et al.* (1972b) from Iceland, although in the latter case the authors claim that their sampling was deliberately biased towards such flows. The Icelandic flows cover the past 12 My. We note however that the number of intermediate flows is especially high (20–30 per cent) for the 35–25 My group, and especially low (3 per cent) for the flows covering the last 5 My. Doell & Cox (1972) in reporting on 1035 lava flows covering much the same time span (the last 5 My) find 3·8 per cent with VGP co-latitudes greater than 40°. Also Aziz-ur-Rahman & McDougall (1973) found only 1 out of 85 lava flows on Norfolk Island in the time range 2 to 3 My had VGP co-latitudes greater than 40°. If all such intermediate poles are due to chance sampling of polarity changes, their observed rate of occurrence over the past 5 My (3–4 per cent) is consistent with the known frequency of about five polarity changes per My (Cox 1969) and an estimated 10^3 to 10^4 years for each change (McElhinny 1971). On the other hand the very much higher rate of occurrence of intermediate flows in the time range 15–60 My is inconsistent with three polarity changes per My estimated from sea-floor spreading over the past 50 My (Heirtzler *et al.* 1968). It appears that the frequency of polarity changes was very much higher during some of the Cenozoic than the sea-floor spreading information predicts. This view is also consistent with

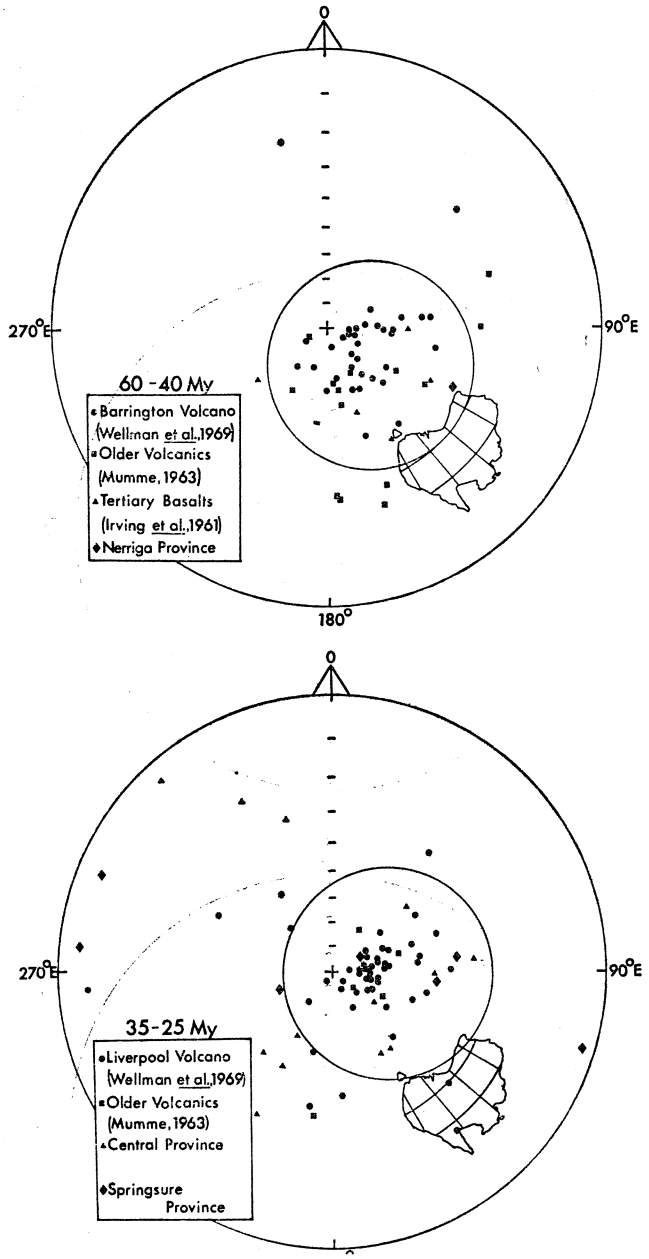


FIG. 2. South polar stereographic projection with pole positions derived from Palaeocene to Oligocene lava flows in eastern Australia. The circle has radius 40° from the mean pole of each group. Poles lying outside this circle are not included in the mean.

some of the detailed observations of Wellman *et al.* (1969). One reason for this could be that there are many relatively short events as has been predicted by Cox (1968) that are as yet undetected. In support of this contention we note that the Reunion event is of very short duration (McDougall & Watkins 1973) and is not observed either in deep-sea sediment cores or from magnetic anomaly profiles. The analyses of Emilia & Hendricks (1972) show that there is evidence for such short events when special techniques are applied to magnetic anomaly profiles. Also it is possible that excursions of the geomagnetic field such as have been observed by Doell & Cox (1972) and Barbetti & McElhinny (1972) occurred more frequently.

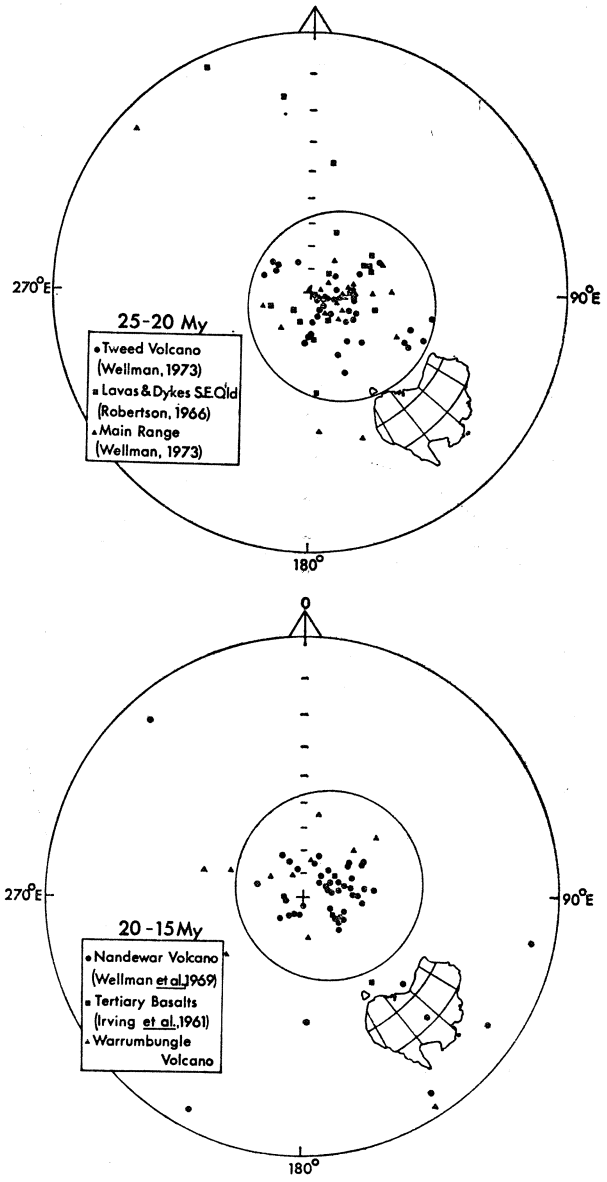


FIG. 3. South polar stereographic projection with pole positions derived from Late Oligocene and Miocene lava flows in eastern Australia. The circle is as described in Fig. 2.

(b) *Polar wander path*

The polar positions observed for each age grouping are illustrated in Figs 2 and 3. The 40° circles within which lie all the poles used in calculating the means are shown. All poles outside these circles are classified as intermediate. The overall mean pole positions as calculated in Table 3 are plotted in Fig. 4; they are significantly different from each other at the 95 per cent probability level. These positions define the apparent polar wander path for Australia during the Cenozoic. For the past 100 My the pole has not followed a simple path from a position just south of Tasmania to the present south pole. Instead the path exhibits a distinct zig-zag with a westerly excursion around 30 My and a knee around 18 My. With the changing configuration of the geomagnetic field during the mid-Tertiary, the fine structure of the pole path is only revealed when sufficiently large numbers of measurements are available. Such a complicated pole path suggests the spreading history between Australia and Antarctica is a little more complicated than that proposed by Le Pichon & Heirtzler (1968) from an initial study of magnetic anomaly profiles. Indeed a later analysis by McKenzie & Sclater (1971) suggests that more than one pole of rotation may have

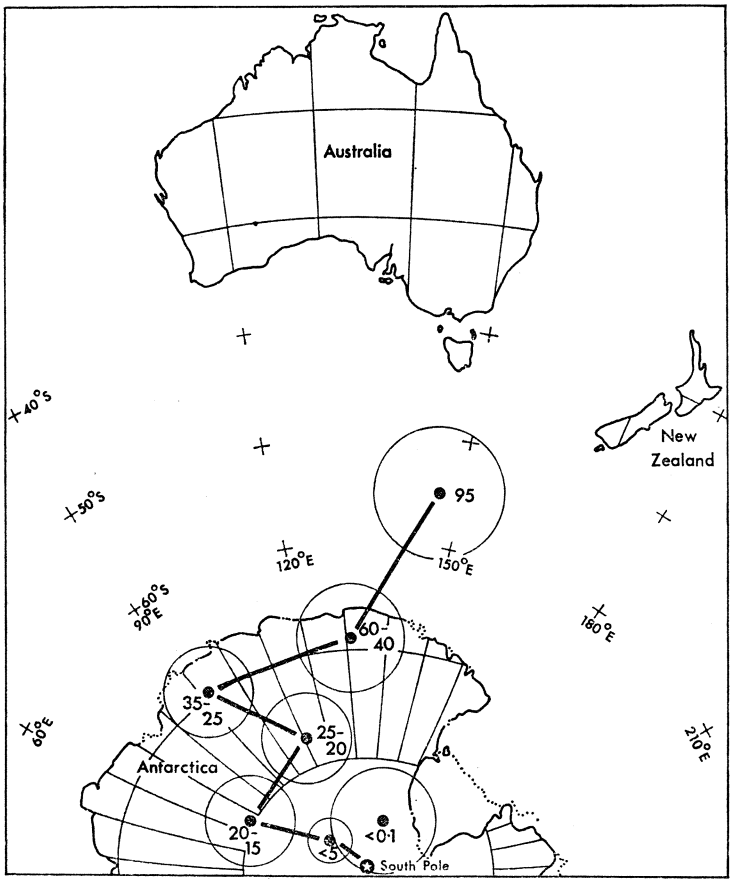


FIG. 4. Apparent polar wander path for Australia since the mid-Cretaceous. The 95-My pole is the mean calculated by McElhinny & Wellman (1969), and the other poles are listed in Table 3.

been involved. This suggestion and the evidence presented by Weissel & Hays (1971) for assymetric sea-floor spreading in this region provides some evidence in support of the general conclusion. McElhinny & Embleton (1973) have described in detail the drift history of eastern Gondwanaland during the late Mesozoic and Cenozoic period.

Research School of Earth Sciences
Australian National University
Canberra, A.C.T.

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